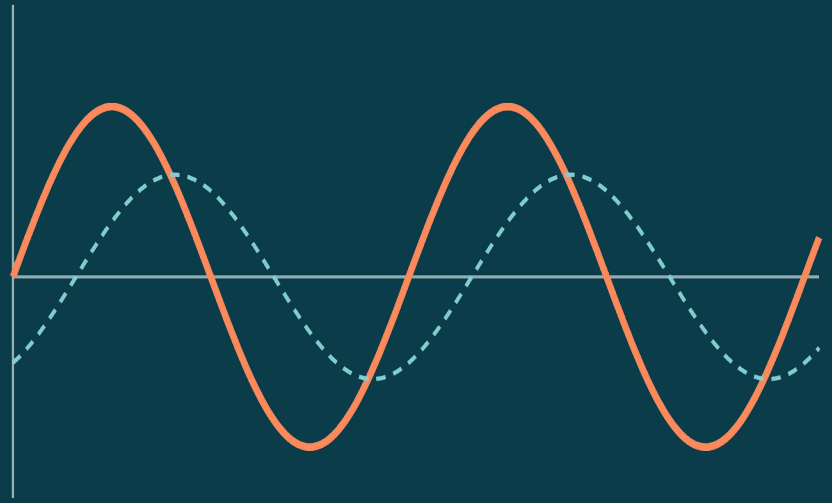
Three-Phase Induction Motor



Basic Electrical and Electronics Engineering

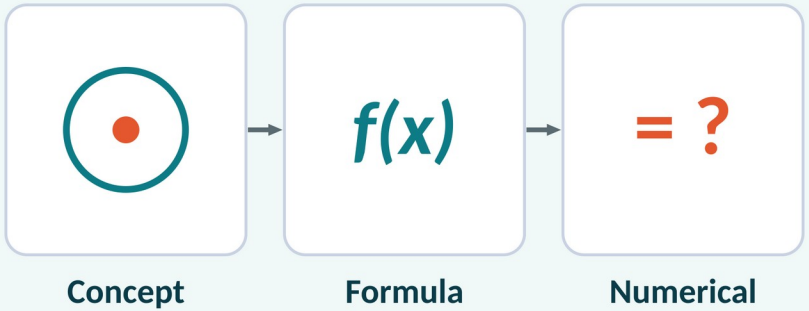
Unit 2: Transformers and Electrical Machines

Topic: Rotating magnetic field, principle, construction, squirrel-cage and wound-rotor types

Prof. Mitesh Astik

Learning Outcomes

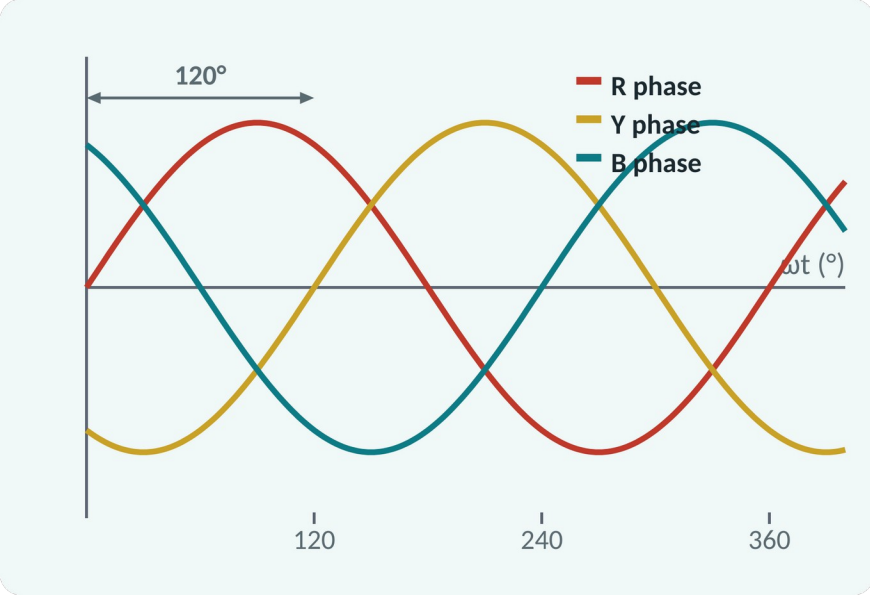
- Explain the rotating magnetic field.
- Describe the working principle and slip.
- Name the parts of an induction motor.
- Compare squirrel-cage and wound-rotor types.



Linked to CO2
Describe construction and operating principle of induction motors.

Recap: What You Already Know

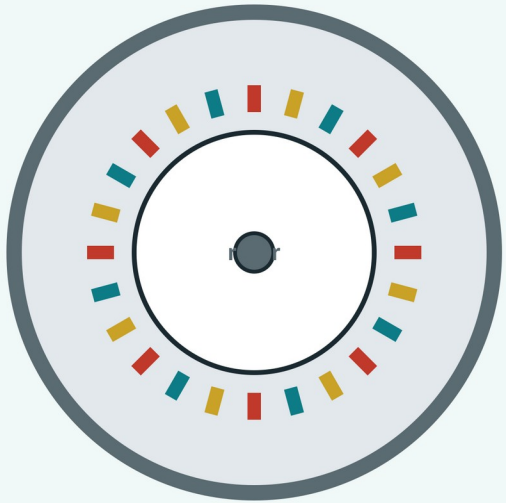
- Three-phase supply: three voltages 120° apart.
- Faraday: moving field induces EMF.
- Lenz: induced current opposes the cause.
- Force on a current-carrying conductor.



VL = √3·Vph (star); phase sequence R–Y–B

Motivation: The Industrial Workhorse

- Over 70 % of industrial motors.
- Pumps, fans, compressors, conveyors.
- Rugged, cheap, little maintenance.
- Runs directly from the three-phase mains.



Stator frame

Laminated stator core

3-phase winding in slots

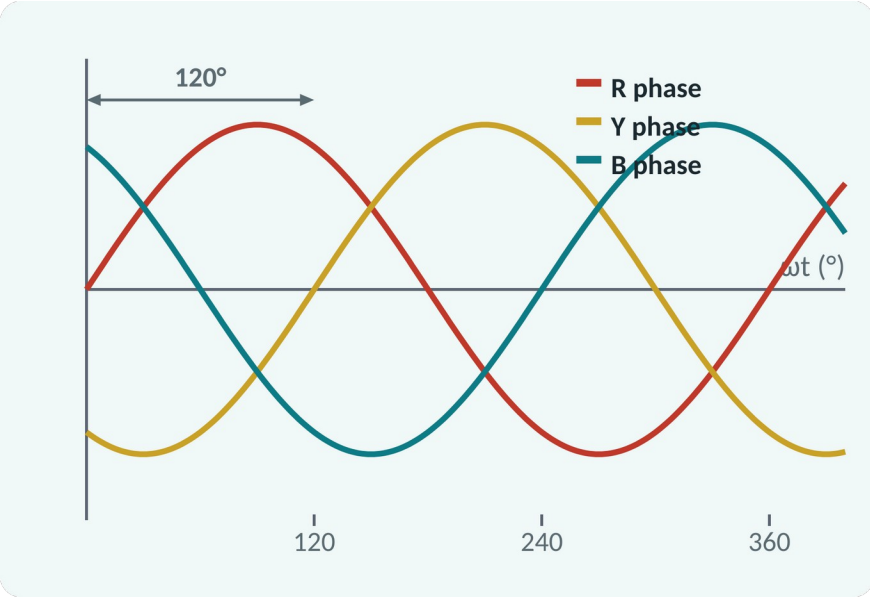
Air gap (0.3–1 mm)

Rotor + shaft

Three-phase induction motor: no commutator, no brushes (squirrel cage)

Three-Phase Supply and Stator Winding

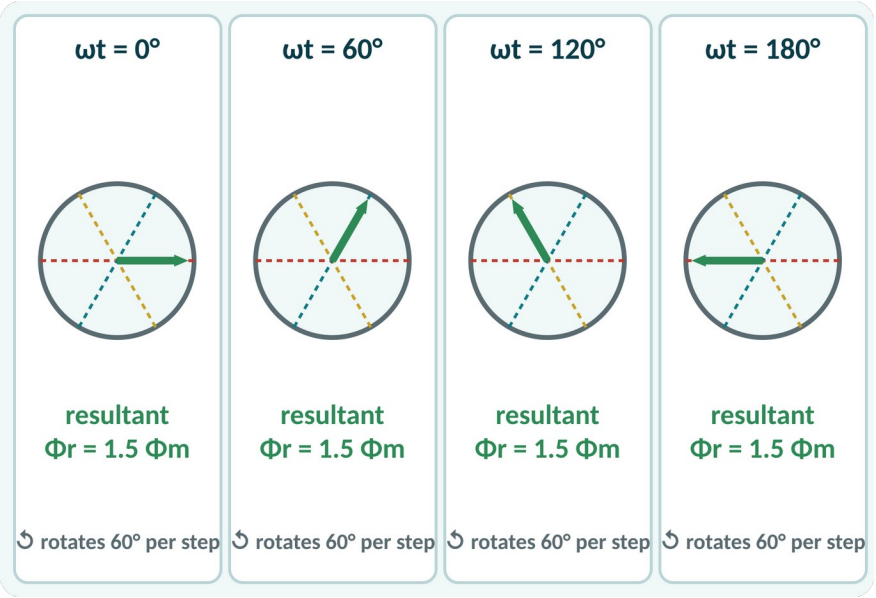
- Three coils placed 120° apart in the stator.
- Each coil carries one phase current.
- Currents are 120° apart in time.
- The three fluxes combine in space.



φR = Φm sin ωt; φY = Φm sin(ωt − 120°); φB = Φm sin(ωt − 240°)

Rotating Magnetic Field

- Add the three fluxes at each instant.
- Resultant has constant magnitude 1.5Φm.
- Its direction advances 60° every 60° of ωt.
- It rotates once per supply cycle (2 poles).
- Reverse two phases to reverse direction.



Φr = 1.5·Φm; speed = synchronous speed Ns

Synchronous Speed

- Speed of the rotating field is Ns.
- Depends on supply frequency and poles.
- More poles: lower speed.
- 50 Hz, 4 poles: 1500 rpm.

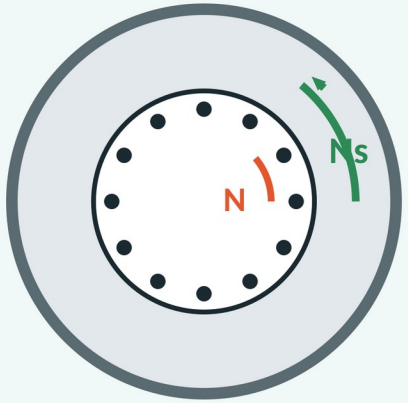
Poles P	Ns at 50 Hz
2	3000 rpm
4	1500 rpm
6	1000 rpm
8	750 rpm
12	500 rpm

Ns = 120 f / P

Ns = 120·f / P (rpm)
f Hz, P number of poles

Principle of Operation

- Rotating field cuts the rotor conductors.
- EMF is induced (Faraday).
- Rotor circuit is closed: current flows.
- Current in the field gives force: torque.
- Rotor follows the field (Lenz) but lags it.



1 RMF cuts the rotor bars

2 EMF induced, rotor current

3 Current in field: force

4 Rotor follows: N < Ns

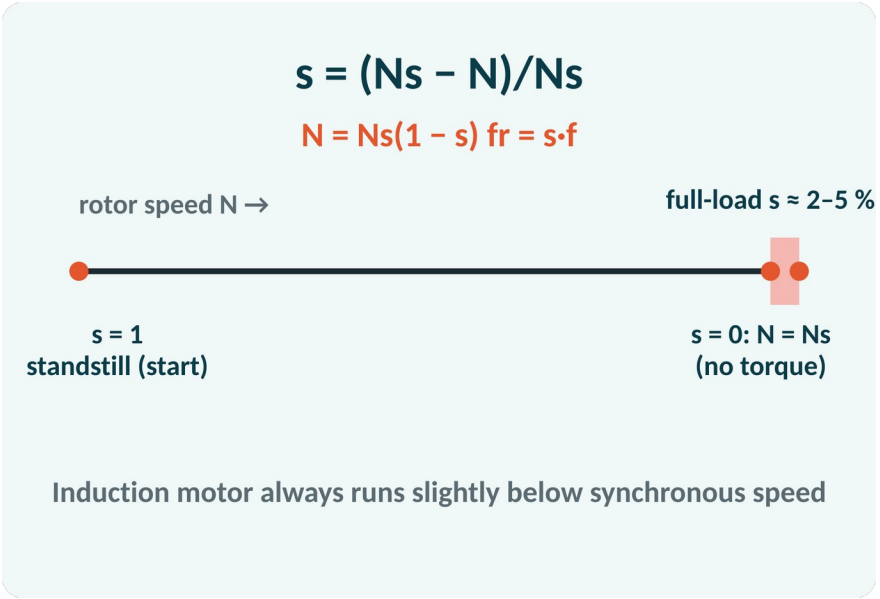
Lenz: rotor chases the field

If N = Ns: no cutting, no torque

Torque exists only if relative speed (Ns − N) ≠ 0

Slip

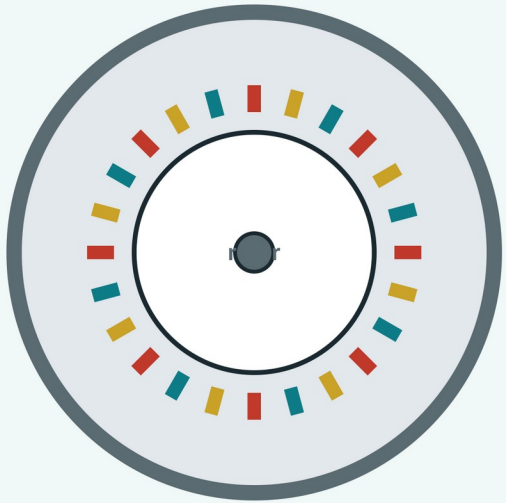
- Slip measures how far N lags Ns.
- Full-load slip is 2–5 %.
- At start N = 0, so s = 1.
- Rotor frequency is f·s: small at full speed.



s = (Ns − N)/Ns; N = Ns(1 − s); fr = s·f

Constructional Features: Stator

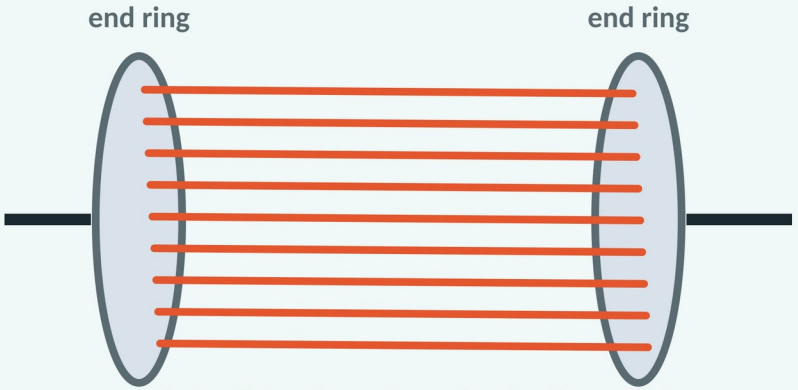
- Frame: cast iron or steel, with cooling fins.
- Stator core: laminated silicon steel.
- Slots hold the three-phase winding.
- Winding terminals in a terminal box.
- Small air gap improves power factor.



Stator winding: three coils displaced 120°; star or delta connected

Squirrel-Cage Rotor

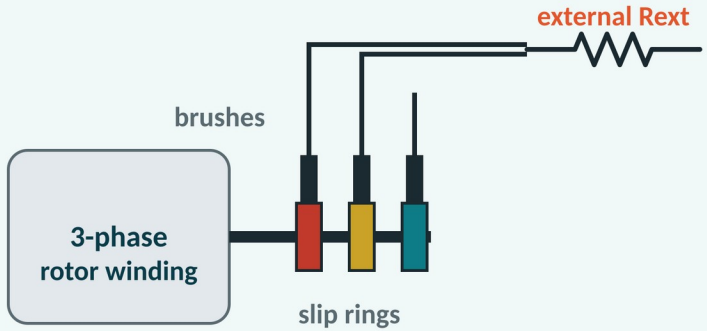
- Laminated core with slots.
- Copper or aluminium bars in the slots.
- Bars joined by end rings: a short-circuited cage.
- No slip rings or brushes.
- Skewed bars reduce noise.



Most common type: simple, rugged and cheap

Wound-Rotor (Slip-Ring) Rotor

- Rotor carries a three-phase winding.
- Winding ends go to three slip rings.
- Brushes connect external resistance.
- High starting torque, low starting current.
- Speed control by rotor resistance.



external resistance → high starting torque, low starting current

cut out as the motor speeds up; speed control possible

Rext cut out gradually as the motor accelerates

Squirrel Cage vs Wound Rotor

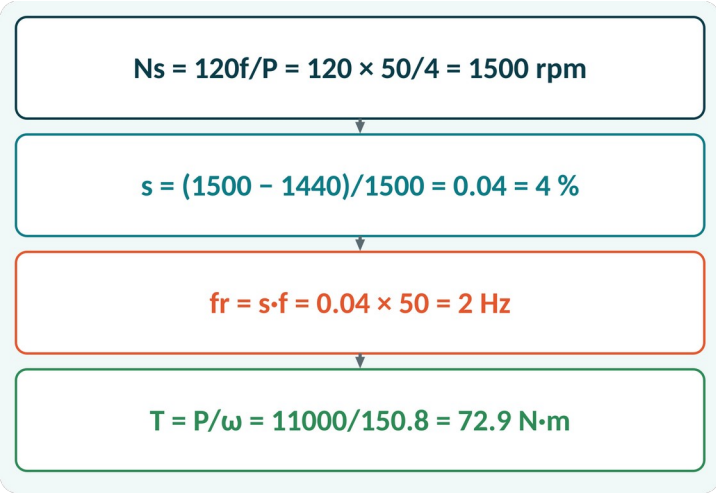
- Cage: cheap, rugged, maintenance-free.
- Wound: slip rings, brushes, costlier.
- Wound rotor gives high starting torque.
- Wound rotor allows speed control.
- Stator is identical in both.

	Squirrel cage	Wound rotor
Rotor	bars + end rings	3-phase winding
Slip rings	none	yes
Cost	low	higher
Maintenance	very little	brushes, rings
Starting torque	moderate	high (with Rext)
Uses	pumps, fans, mills	cranes, hoists

Both run on the same rotating-field principle

Solved Example 1: Slip and Torque

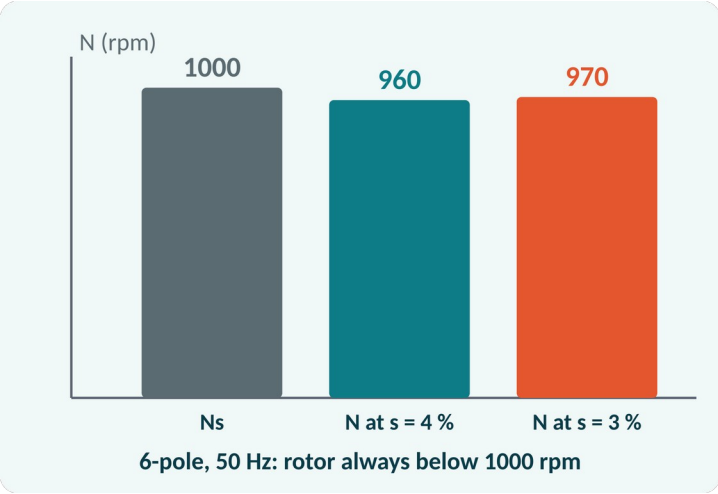
- A 4-pole, 50 Hz induction motor runs at 1440 rpm and delivers 11 kW. Find Ns, slip, rotor frequency and shaft torque.
- 1. Ns = 120 × 50/4 = 1500 rpm.
- 2. s = (1500 − 1440)/1500 = 0.04 (4 %).
- 3. fr = s·f = 2 Hz.
- 4. ω = 2π × 1440/60 = 150.8 rad/s.
- 5. T = P/ω = 11000/150.8 = 72.9 N·m.



Answer: Ns = 1500 rpm, slip 4 %, fr = 2 Hz, T = 72.9 N·m

Solved Example 2: Speed from Slip

- A 6-pole, 50 Hz induction motor has full-load slip 4 %. Find the speed, and the speed if the load is reduced so that slip is 3 %.
- 1. Ns = 120 × 50/6 = 1000 rpm.
- 2. N = Ns(1 − s) = 1000 × 0.96 = 960 rpm.
- 3. At s = 3 %: N = 1000 × 0.97 = 970 rpm.
- 4. Rotor frequency at full load = 0.04 × 50 = 2 Hz.



6-pole, 50 Hz: rotor always below 1000 rpm

Answer: 960 rpm at full load, 970 rpm at lighter load; Ns = 1000 rpm

Common Mistakes and Exam Tips

Common mistakes

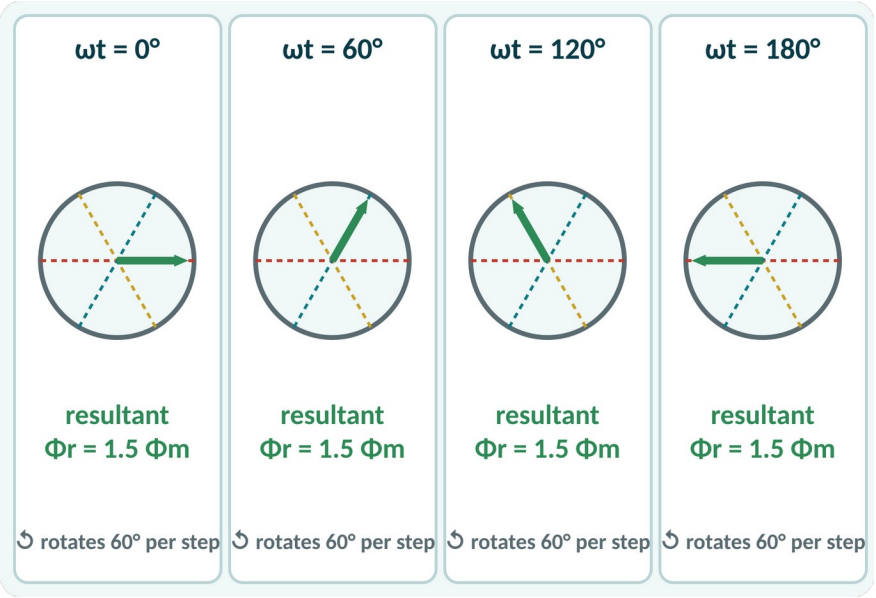
- Using N = Ns (ignoring slip).
- Using P as the number of pole pairs in Ns = 120f/P.
- Saying the rotor runs at synchronous speed.
- Mixing slip as % and as a fraction.
- Forgetting to swap two phases to reverse direction.

Exam tips

- Ns = 120f/P with P = number of poles.
- Slip as a fraction in formulas.
- Rotor frequency fr = s·f.
- Torque needs ω in rad/s.

Summary

- Three-phase currents produce a rotating field of constant magnitude.
- Ns = 120f/P.
- Rotor follows the field with slip s = (Ns − N)/Ns.
- Stator: laminated core with 3-phase winding.
- Rotor: squirrel cage (simple) or wound (slip rings).



Ns = 120f/P; N = Ns(1 − s); fr = s·f

Quick Quiz

Q1. Synchronous speed for 50 Hz, 6 poles:
(a) 500 (b) 1000 (c) 1500 (d) 3000 rpm

Ans: (b)

Q2. Slip at standstill is:
(a) 0 (b) 0.5 (c) 1 (d) ∞

Ans: (c)

Q3. To reverse the direction of rotation:
(a) swap two supply leads (b) swap all leads (c) change frequency (d) open one phase

Ans: (a)

Q4. The rotor of a squirrel-cage motor has:
(a) slip rings (b) bars and end rings (c) a commutator (d) permanent magnets

Ans: (b)

Q5. N = 1450 rpm, Ns = 1500 rpm gives slip:
(a) 3.3 % (b) 5 % (c) 33 % (d) 0.33 %

Ans: (a)

References

1. **Hughes Edward, Electrical Technology**
   10th Ed., Pearson Education, 2010 (Units 1, 2 textbook)

2. **H. Cotton, Electrical Technology**
   7th Ed., CBS Publishers, 2005 (reference)

Next lecture: Unit 3: semiconductor diode and rectifiers